

SECTION 081100 - STEEL FRAMES

PART 1 - GENERAL

1.1 SUMMARY

- A. Provide steel frames in the work of this section.
- B. Refinish existing hollow metal doors and frame. Comply with requirements specified in Division 02, Section "Selective Demolition" and Division 09 Section "Painting".

1.2 QUALITY ASSURANCE:

- A. Manufacturer: Steel frames for fire rated openings shall be manufactured by a firm whose units are inspected, tested, listed and labeled by UL for single point hardware.
- B. Safety Glazing Materials: Comply with the requirements of 16 CFR 1201 of the Consumer Products Safety Act with respect to glass and glazing provided in conjunction with steel frames. Refer to Section "Glazing".
- C. STC Test Reports: Certified test reports from an independent testing laboratory showing STC ratings for acoustical door assemblies.
- D. Reference Standards: Provide frames complying with requirements of the following standards:
 - 1. ANSI/SDI A250.8 "Recommended Specifications for Standard Steel Doors and Frames" and as specific.
 - 2. DHI "Recommended Locations for Finish Hardware."

1.3 SUBMITTALS

- A. Product Data for each type of frame required, including details of construction, materials, dimensions, hardware preparation, core, and finishes.
- B. Shop Drawings: Submit shop drawings for each typical frame opening. Show typical elevations and details, details of joints and connections to other work, description of anchorage and accessory items and location of reinforcement and installation requirements for finish hardware.
 - 1. Indicate coordination of glazing frames and stops with glass and glazing requirements.
- C. Schedule: Door and frame Schedule utilizing reference numbers for details and openings as shown.
- D. Sustainable Design Submittals:
 - 1. Product Data: For products having recycled content, documentation indicating percentages by weight of postconsumer and preconsumer recycled content. Include statement indicating cost for each product having recycled content.

- a. Product Data: Indicate location of material manufacturer for regionally manufactured materials. Include statement indicating cost for each regionally manufactured material.
 - 1) Include statement indicating distance from manufacturer to Project for each regionally manufactured material.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Basis of Design: Frames specified herein are as manufactured by Steelcraft. Equivalent Products by one of the following manufacturers will be accepted subject to compliance with requirements of these specifications.
 1. **Ceco Door Products**; an ASSA ABLOY Group Company.
 2. Curries; and ASSA ABLOY Group Company
 3. Fenestra Corp.
 4. Pioneer Industries.
 5. Republic Builders Products.
 6. Steelcraft; and Ingersoll Rand Company.
- B. Materials
 1. Recycled Content of Steel Products: Postconsumer recycled content plus one-half of preconsumer recycled content not less than 45 percent.
 2. **Metallic Coated Sheet Steel**: ASTM A653 and ASTM A924, Commercial Steel (CS), **Type B, with an A60 zinc iron alloy (galvannealed) coating**, mill phosphatized; stretcher leveled standard of flatness for frames. Factory prepare metallic coated sheet steel components to receive finish painting in the field.
 3. Reinforcing, Supports and Anchors: Structural steel, ASTM A36, or sheet steel, ASTM A366 cold rolled or ASTM A569 hot rolled and ASTM A568; **but not less than 16 gauge**; hot dip galvanize after fabrication; comply with ASTM A153, **Class B**.
 4. Frame Filler: Heat-retarding mineral fiber insulating material.
 5. Inserts, Bolts, and Fasteners: Manufacturer's standard units.
 6. Power-Actuated Fasteners in Concrete: From corrosion-resistant materials.
 7. Grout: ASTM C476, except with a maximum slump of 4 inches, as measured according to ASTM C143.
 8. Shop Primer: A baked-on primer compatible with the respective specified finish paint and complying with the test procedure and acceptance criteria of ANSI A250.10
 9. Glazing: Division 08 Section "Glazing."
 10. Bituminous Coating: Cold-applied asphalt mastic, compounded for 15 mil dry film thickness per coat.

2.2 FABRICATION

- A. Fabricate steel frame units to be rigid, neat in appearance, and free from defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for metal thickness. Where practical, fit and assemble units in manufacturer's plant. To ensure proper assembly at Project site, clearly identify work that cannot be permanently factory assembled before shipment. Comply with ANSI/SDI A250.8 requirements.

- B. Frames: Provide metal frames according to ANSI/SDI 100, of types and styles shown on drawings and schedules.
1. Fabricate new steel frames of 16 gage metallic coated steel sheet.
 2. Construction: Full profile welded.
 3. Fabricate frames for fire rated openings and frames for new partitions with mitered or coped and continuously welded corners.
 4. Floor Anchors: Weld anchors to bottoms of jambs with at least four spot welds per anchor; however, for slip-on drywall frames, provide anchor clips or countersunk holes at bottoms of jambs.
- C. Sidelite and Transom Bar Frames: Provide closed tubular members with no visible face seams or joints, fabricated from same material and thickness as frame. Fasten members at crossings and to jambs by butt welding.
1. Construction: Full profile welded.
 2. Exposed Finish: Prime.
- D. Borrowed Lite Frames: Provide borrowed lite frames of same material, thickness, construction types, and finish as frames.
- E. Fire Rated Assemblies: Fabricate fire rated steel frames in accordance with requirements of NFPA 80, and listed and labeled by a qualified testing agency acceptable to authorities having jurisdiction for fire protection ratings and temperature rise limits indicated, based on testing at positive pressure according to NFPA 252 or UL 10C.
1. Smoke and Draft Control Assemblies: Provide an assembly with gaskets listed and labeled for smoke and draft control by a qualified testing agency acceptable to authorities having jurisdiction, based on testing according to UL 1784 and installed in compliance with NFPA 105.
 2. Fire Rated, Borrowed Lite Assemblies: Complying with NFPA 80 and listed and labeled by a testing and inspecting agency acceptable to authorities having jurisdiction for fire protection ratings indicated, based on testing according to NFPA 257 or UL 9.
- F. Tolerances: Comply with SDI 117 "Manufacturing Tolerances Standard Steel Doors and Frames."
- G. Hardware Preparation: Prepare frames to receive mortised and concealed hardware according to final door hardware schedule and templates provided by hardware supplier. Comply with applicable requirements of SDI A250.6 and ANSI A115 Series specifications for frame preparation for hardware.
1. For concealed overhead door closers, provide space, cutouts, reinforcing, and provisions for fastening in top rail of doors or head of frames, as applicable.
 2. Reinforce frames to receive surface applied hardware. Drilling and tapping for surface applied hardware may be done at Project site.
 3. Locate hardware as indicated on Shop Drawings or, if not indicated, according to the Door and Hardware Institute's (DHI) "Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames."
- H. Finishes: Thoroughly clean metal surfaces of loose scale, shavings, filings, dirt and other deleterious materials by use of wire brushes or other effective means. Remove grease and oil

by one of the methods specified in SSPC-SP-1 "Solvent Cleaning". Fill as required to fill seams in edges.

1. Chemically treat surfaces of metallic coated sheet steel frames with phosphate compound to assure maximum paint adhesion. Treat and prime paint metallic coated sheet steel frames prior to finish coats of paint. Comply with requirements of Division 09 Section "Painting"

PART 3 - EXECUTION

3.1 INSTALLATION

- A. General: Install steel frames and accessories according to final shop drawings, manufacturer's data, and as specified.
- B. Placing Frames: Comply with provisions of SDI A250.11, unless otherwise indicated. Set frames accurately in position, plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is completed, remove temporary braces and spreaders, leaving surfaces smooth and undamaged.
 1. At fire rated openings, install frames according to NFPA 80.
 2. Where frames are fabricated in sections because of shipping or handling limitations, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces.
 3. Install frames with removable stops located on secure side of opening.
 4. Install door silencers in frames before grouting.
 5. Remove temporary braces necessary for installation only after frames have been properly set and secured.
 6. Check plumb, square, and twist of frames as walls are constructed. Shim as necessary to comply with installation tolerances.
 7. Field apply bituminous coating to backs of frames that will be filled with grout containing antifreezing agents.
 8. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with postinstalled expansion anchors.
 - a. Floor anchors may be set with power-actuated fasteners instead of postinstalled expansion anchors if so indicated and approved on Shop Drawings.
 9. Metal Stud Partitions: Solidly pack mineral fiber insulation inside frames.
 10. Masonry and Concrete Walls: Solidly fill space between frames and concrete with mineral fiber insulation.
 11. In-Place Concrete or Masonry Construction: Secure frames in place with postinstalled expansion anchors. Countersink anchors, and fill and make smooth, flush, and invisible on exposed faces.
 12. In-Place Metal Stud Partitions: Secure slip-on drywall frames in place according to manufacturer's written instructions.
- C. Glazing: Comply with installation requirements in Division 08 Section "Glazing" and with hollow metal manufacturer's written instructions.
 1. Secure stops with countersunk flat or oval-head machine screws spaced uniformly not more than 9 inches o.c. and not more than 2 inches o.c. from each corner.

- D. Apply hardware in accordance Division 08 Section "Door Hardware" of these Specifications. Drill and tap for machine screws as required. Do not use self-tapping sheet metal screws. Anchor panels in place with concealed fasteners.
- E. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow metal work that is warped, bowed, or otherwise unacceptable

END OF SECTION

SECTION 081416 - WOOD DOORS

PART 1 - GENERAL

1.1 SUBMITTALS

- A. Product Data: For each type of door. Include factory finishing specifications.
- B. Sustainable Design Submittals:
 - 1. Product Certificates: For products and materials required to comply with requirements for regionally manufactured and regionally extracted and manufactured materials.
 - a. Include statement indicating cost for each regionally manufactured material.
 - b. Include statement indicating location of manufacturer and distance to Project for each regionally manufactured material.
 - c. Include statement indicating location of manufacturer and point of extraction, harvest, or recovery for each raw material used in regionally extracted and manufactured materials. Indicate distance to Project and fraction by weight of each regionally manufactured material that is regionally extracted.
 - 2. Certificates: Chain-of-custody certificates indicating that flush wood doors comply with forest certification requirements. Include documentation that manufacturer is certified for chain of custody by an FSC-accredited certification body. Include statement indicating cost for each certified wood product.
 - 3. Product Data: For adhesives and composite wood products, documentation indicating that product contains no urea formaldehyde.
- C. Shop Drawings: Indicate location, size, and hand of each door; elevation of each kind of door; construction details not covered in Product Data
- D. Samples: For factory finished doors.
 - 1. Provide two samples of wood veneered wood door 4 inch x 4 inch x full width, with stain and finish to match existing. Approved sample shall serve as a quality control sample for wood veneered doors.
- E. Quality Standard Compliance Certificates: AWS Quality Certification Program Certificates.
- F. Manufacturer/Vendor Qualifications

1.2 QUALITY ASSURANCE

- A. Manufacturer Qualifications: A qualified manufacturer that is certified for chain of custody by an FSC-accredited certification body and is a certified participant in the AWI's Quality Certification Program.
- B. Vendor Qualifications: A vendor that is certified for chain of custody by an FSC-accredited certification body

1.3 PROJECT CONDITIONS

- A. Maintain temperature and relative humidity complying with referenced standards and requirements specified in Division 06, Section "Interior Architectural Woodwork", during fabrication, storage, handling and installation of wood doors.

1.4 WARRANTY

- A. Special Warranty: Manufacturer's standard form, signed by manufacturer, Installer, and Contractor, in which manufacturer agrees to repair or replace doors that are defective in materials or workmanship, have warped (bow, cup, or twist), or show telegraphing of core construction in face veneers.
 - 1. Warranty shall also include installation and finishing that may be required due to repair or replacement of defective doors.
 - 2. Warranty shall be in effect during the following period of time from date of Substantial Completion: Solid Core Interior Doors: Life of installation.

PART 2 - PRODUCTS

2.1 MATERIALS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Algoma Hardwoods, Inc.
 - 2. Eggers Industries.
 - 3. Marshfield Door Systems, Inc.
 - 4. Mohawk Doors; a Masonite company.
- B. Quality Standard: In addition to requirements specified, comply with AWI's, AWMAC's, and WI's "AWS."
 - 1. Provide AWI Quality Certification Labels indicating that doors comply with requirements of grades specified.
- C. Certified Wood: Flush wood doors shall be certified as "FSC Pure" or "FSC Mixed Credit" according to FSC STD-01-001, "FSC Principles and Criteria for Forest Stewardship," and to FSC STD-40-004, "FSC Standard for Chain of Custody Certification."
- D. Low Emitting Materials: Fabricate doors with adhesives and composite wood products that do not contain urea formaldehyde.
- E. Cores:
 - 1. Structural Composite Lumber: WDMA I.S.10.
 - a. Screw Withdrawal, Face: 700 lbf.
 - b. Screw Withdrawal, Edge: 400 lbf.

2. **Mineral Core:** Noncombustible mineral product complying with requirements of referenced quality standard and testing and inspecting agency for fire protection rating indicated.
 3. Blocking: Provide composite blocking with improved screw-holding capability approved for use in doors of fire protection ratings indicated as needed to eliminate through-bolting hardware.
 4. Edge Construction: At hinge stiles, provide laminated edge construction with improved screw-holding capability and split resistance. Comply with specified requirements for exposed edges.
- F. Hardwood veneers complying with AWS Section 4, Premium Grade, free from sapwood, knots, and other defects. Where solid wood is required, provide wood of same species, matching adjoining veneers unless otherwise shown or specified and free from cat's eyes, bird's eyes, burls, curls or cross grains.
- G. Flush Wood Door Construction: AWS Section 9, Premium Grade, PC-5. Stiles and rails bonded to core and entire unit abrasive planed before veneering. Provide three ply particle board cores, except at fire rated doors. Faces are bonded to core using a hot process.
1. Vertical Stiles: Minimum 1-3/8 in. wide after trimming, solid hardwood or engineered hardwood construction without finger joints, consisting of 5/8 in. thick outerband (specie matching or compatible with face veneer) and 3/4 in. thick mill option hardwood innerband.
 2. Top and Bottom Rails: Minimum 1-3/8 in. wide after trimming, solid hardwood or engineered hardwood construction without finger joints. For doors scheduled to receive surface mounted overhead closers, utilize 5 in. solid wood rails so that doors are not through-bolted.
 3. Crossbands: Full width of door with grain running horizontally, tapeless spliced without voids.
 4. Face Veneer: **WDV-xx** specified in the Material and Finish Schedule on Drawings. Laminated to crossband and edges.
 5. Bonding: Provide wood blocking in particleboard core doors needed to eliminate through-bolting hardware
 - a. Core to stiles, rails and blocking utilizing adhesive.
 - b. Face veneers and crossbands to core and edges utilizing adhesive by hot plate process.
 6. **Cores: Provide structural composite lumber cores instead of particleboard cores at full height pivot doors, and doors to receive exit devices.**
- H. Stile and Rail Door Construction: Provide stile and rail glass doors complying with AWS Section 9, and other requirements specified.
1. Grade: **Premium.**
 2. Finish: **Opaque.**
 3. Stile and Rail Construction: Structural composite lumber, or veneered edge and end glued lumber.
 4. Widths: As indicated
 5. Assemble interior doors, including components, with either dry-use or wet-use adhesives complying with ASTM D5572 for finger joints and with ASTM D5751 for joints other than finger joints.

- I. Fire Rated Flush Wood Door Construction: AWS Premium Grade, FD-5, or HPDL-5. Construct in accordance with requirements of NFPA 80 listed and labeled by a qualified testing agency, for fire protection ratings indicated, based on testing at positive pressure according to NFPA 252 or UL 10C. Provide fire rated label on each door. Comply with requirements specified herein and the following minimum requirements:
1. Core: Incombustible, asbestos-free mineral composition, engineered to meet labeling requirements.
 2. Vertical Stiles: Provide balanced construction by furnishing stile edge screw holding reinforcement to both vertical stiles as standard with the manufacturer and complying with the following:
 - a. Stile edge screw withdrawals when tested in accordance with ASTM D1037 shall exceed 740 lbs.
 - b. Stile edge split resistance when tested in accordance with ASTM D143 (modified) shall exceed 750 lbs.
 - c. Cycle/Slam: Not less than 500,000 cycles with no loosening of hinge screws or other visible signs of failure when tested in accordance with the requirements of ANSI A151.1.
 3. Top and Bottom Rails: Top rails minimum 1/2 in. wide fire retardant treated wood after trimming, bottom rails minimum 1-1/2 in. wide after trimming with finger joints. Provide additional reinforcing as required by specified hardware. At Doors to receive surface mounted overhead closers, utilize 5 in. solid wood rails so that doors are not through-bolted.
 4. Crossbands: Full width of door with grain running horizontally, tapeless spliced without voids.
 5. Bonding: Manufacturer's standard to achieve specified rating.
 6. Blocking
 - a. Provide 5 in. blocking at top rail for surface door closers.
 - b. Provide 5 in. x 18 in. blocking for locksets and exit devices.
 - c. Provide blocking for flushbolts and vertical rod exit device latches.
 7. Construct fire rated wood doors with material containing no added urea formaldehyde.
 8. Edge Construction: Provide edge construction with intumescent seals concealed by outer stile. Comply with specified requirements for exposed edges.
 9. Pairs: Provide fire retardant stiles that are listed and labeled for applications indicated without formed steel edges and astragals. Provide stiles with concealed intumescent seals. Comply with specified requirements for exposed edges.
- J. Smoke and Draft Control Door Assemblies: Listed and labeled for smoke and draft control, based on testing according to UL 1784.
- K. Doors for Transparent Finish: Premium grade finished as follows:
1. Provide premium grade finish equal AWI's, "Architectural Woodwork Standards" System TR-5, conversion varnish, satin. Match existing wood veneer and finish for staining requirements.
 2. Provide door hardware specified in the door and hardware schedule.
- L. Doors for Opaque Finish: Custom grade, with faces of closed grain hardwood, finished as follows:
1. Shop primed. Finish doors in the field, as specified in Section "Interior Painting".

2. Provide premium grade finish equal AWI's, "Architectural Woodwork Standards" System 5, conversion varnish, satin. Match Architect's approved sample for staining requirements.
3. Provide door hardware specified in the door and hardware schedule.
4. Provide opaque finished sliding wood pocket doors with hardware specified in the door and hardware schedule.

2.2 FABRICATION

- A. Factory fit doors to suit frame opening sizes indicated. Comply with clearance requirements of referenced quality standard for fitting unless otherwise indicated.
 1. Comply with NFPA 80 requirements for fire rated doors.
- B. Factory machine doors for hardware that is not surface applied. Locate hardware to comply with DHI-WDHS-3. Comply with final hardware schedules, door frame Shop Drawings, DHI A115-W series standards, and hardware templates.
 1. Coordinate with hardware mortises in metal frames to verify dimensions and alignment before factory machining.
- C. Openings: Cut and trim openings through doors in factory.
 1. At glazed openings, provide solid wood moldings to support glass and glazing, with secure side removable for Reglazing. Factory install glazing in doors complying with requirements specified in Division 08, Section "Glazing". Install glass using manufacturer's standard glazing tapes and secure in place with removable wood moldings. Miter joint wood moldings at corner joints
- D. Fabricate fire rated doors and frames identical to UL tested assemblies for fire rating indicated.

2.3 FACTORY FINISHING

- A. General: Comply with referenced quality standard for factory finishing. Complete fabrication, including fitting doors for openings and machining for hardware that is not surface applied, before finishing.
 1. Finish faces and all four edges, edges of cutouts, and mortises. Stains and fillers may be omitted on top and bottom edges, edges of cutouts, and mortises.
 2. Transparent Finish:
 - a. Grade: Premium.
 - b. Finish: AWI's, AWMAC's, and WI's "Architectural Woodwork Standards" [System 5, conversion varnish] [System 9, UV curable, acrylated epoxy, polyester, or urethane] [System 10, UV curable, water based] [or] [System 11, catalyzed polyurethane].
 - c. Staining: [Match Architect's sample] [None required].
 - d. Effect: [Open-grain finish] [Filled finish] [Semi-filled finish, produced by applying an additional finish coat to partially fill the wood pores].
 - e. Sheen: [Satin] [Semigloss].
- B. Repair of Factory Finishes: Interior field applied repair and touch-up paint and coatings shall meet the following VOC limits:

1. Prime Paint: 50 grams/Liter for flat paint; 50 grams/Liter for non-flat paint.
2. Lacquer - Clear or Pigmented: 275 grams/Liter.
3. Wood sealer: 200 grams/Liter.
4. Wood stain: 100 grams/Liter.
5. Opaque Finish:
 - a. Grade: Premium.
 - b. Finish: AWI's, AWMAC's, and WI's "Architectural Woodwork Standards"
[System 5, conversion varnish] [System 9, UV curable, acrylated epoxy, polyester, or urethane] [System 10, UV curable, water based] [or] [System 11, catalyzed polyurethane] <Insert finish designation>.
 - c. Color: [Match Architect's sample]<insert color>.
 - d. Sheen: [Satin][Semigloss][Gloss].

PART 3 - EXECUTION

3.1 PREPARATION

- A. Condition doors to average prevailing humidity in installation area prior to hanging. Field cutting, trimming, fitting and machining of prefinished doors and panels will not be permitted.

3.2 EXAMINATION

- A. Examine doors and installed door frames before hanging doors.
 1. Verify that frames comply with indicated requirements for type, size, location, and swing characteristics and have been installed with level heads and plumb jambs.
 2. Reject doors with defects.
 3. Proceed with installation only after unsatisfactory conditions have been corrected.

3.3 INSTALLATION

- A. Install doors to comply with manufacturer's written instructions, referenced quality standard, and as indicated.
 1. Install doors in required openings as shown with clearances conforming to AWS. Install flush panels with concealed fasteners. Provide pilot holes of proper size for screws into particle board core doors. Use stainless steel sheet metal screws into particle board.
 - a. Provide 1/8 in. at jambs and heads; 1/16 in. per leaf at meeting stiles for pairs of doors; and 1/8 in. from bottom of door to top of decorative floor finish or covering.
 - b. Where threshold is shown or scheduled, provide 1/4 in. clearance from bottom of door to top of threshold. Bevel non rated doors 1/8 in. in 2 in. at lock and hinge edges.
 - c. Comply with NFPA 80. Bevel non-fire rated doors 1/8 inch in 2 inches at lock and hinge edges, trim stiles and rails only to extent permitted by labeling agency.
 2. Install fire rated doors in corresponding fire rated frames in accordance with requirements of the UL Label, NFPA No. 80 and local authorities having jurisdiction.
 3. Install smoke and draft control doors according to NFPA 105.

- B. Job Fitted Doors: Align and fit doors in frames with uniform clearances and bevels as indicated below; do not trim stiles and rails in excess of limits set by manufacturer or permitted for fire rated doors. Machine doors for hardware. Seal edges of doors, edges of cutouts, and mortises after fitting and machining.
- C. Factory Fitted Doors: Align in frames for uniform clearance at each edge.
- D. Apply hardware in accordance with hardware manufacturer's instructions and Division 08 Section "Door hardware". Adjust door installation to provide uniform clearance at head and jambs, and to contact stops uniformly. Remove and replace doors which are found to be warped, bowed or otherwise damaged and cannot be properly fitted in openings.
- E. Adjust and Clean: Rehang and replace doors which do not swing or operate properly. Marred finishes shall be touched-up to perfectly match adjacent surfaces to the satisfaction of the Architect or unit shall be replaced

END OF SECTION